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Open Ideas

- One idea I never had time to implement was looking at collaboration trajectories across topics (keywords on paper) - i.e. analysing the distributions of topics that a typical computer science researcher is most likely to view for collaborations given their previous topics. This could be a nice extension here for the separate PMFI spaces if they can be labelled for a more fixed ontology traversal across the embedding space
- Collaborations between researchers can take the displacement from each individuals' previous trajectories, problems or interpretations to their collaborative paper's trajectory or spaces to analyse the interdisciplinarity and how that affects impact - both having short displacements, long or mixed displacements and whether that affects impact more or less. Seeing whether the geometry (S) is more similar to one researcher or the other - author topic influence.
- Transition distributions using KNN, radius tolerance
- FUTURE Research - POMDP as synthetic processes to explore the knowledge space
- Do aggregate trajectories/transitions fit a gravity model — jumps more frequent between dense regions and decay with distance.

Project Resources

files/assets	
 File/URL	 Description
Modeling Scientific Discovery as Discourse Trajectories in Epistemic Landscapes	Overleaf Next Paper working draft.
Towards a mathematical theory of science	Overleaf Formulation for the maths behind the paper.
Remote Server	Ssh -L 8050:localhost:8050 icorreia@erdos.ex.ac.uk -p 9022 Password in vault. Data/prototype stored here initially with repo using server resources.
Azure Storage 1TB	Ask Owen for AWS CLI access credentials if needed
 File	
URL	

external resources			
 File/URL	 Description	 Notes	 Type
Global Research Space	Latest 10M papers from OpenAlex and generated embeddings using SPECTER 2 on titles and abstracts. Reduced dimensionality with UMAP, then applied Voronoi partitioning on density peaks to create distinct	Live - Adds new papers monthly	Website ▾

external resources

📄 File/URL	T T Description	T T Notes	🔍 Type
	semantic neighborhoods.		
Charting Mobility Patterns in the Scientific Research Landscape	1.45M arXiv pre-prints embedded into a 2D space via tSNE on 175 binary sub-field tags. Researcher publication histories form trajectories; jump distances follow a power-law (rarely long-range). Aggregate flows fit a gravity model — jumps more frequent between dense regions and decay with distance. Explorer vs. Exploiter Dichotomy Trajectories split into two types: <i>explorers</i> who range widely across fields, and <i>exploiters</i> who stay within a narrow region. Explorers are more associated with disruptive, cross-field work.	argues that human mobility formalisms (gravity model, trajectory classification) transfer cleanly to the scientific knowledge landscape as a framework for studying knowledge evolution	Paper ▾
OpenAlex	316 million scholarly works, linking them to authors, institutions, funders, and more	Pre-2010 missing many attributes - such as citation count by year Domain	Data ▾

external resources

 File/URL	 Description	 Notes	 Type
		categorisation is estimated ~90% accurate, e.g. Quantum physics appears in Computer science occasionally	
PubMed XML Abstracts	Baseline Data ----- NLM produces a baseline set of PubMed citation records in XML format for download on an annual basis. The annual baseline is released in December of each year. The complete baseline consists of files pubmed26n0001.xml through pubmed26n1334.xml.	Documentation here	Data ▾
A PubMed-Scale Dataset of Structured Biomedical Abstracts	Structured PubMed, corpus of section-labeled abstracts (PubMed database), 23.2 million articles. 5.9 million author-structured abstracts parsed from official XML, and 17.2 million originally unstructured abstracts structured via a LLM pipeline	Very Recent (9th June 2026) BACKGROUND: Context leading up to the gap in knowledge. - OBJECTIVE: Specifically the statement of intent (e.g., "We aimed to...", "This study evaluates..."). If inextricably linked to Background, you may	Paper ▾

external resources

 File/URL	 Description	 Notes	 Type
		split the text at the transition. - METHODS: Technical steps, participant criteria, and analysis. - RESULTS: The specific data found. - CONCLUSIONS: The final takeaway and interpretation	
Structured PubMed	As above.	Dataset for above paper	Data ▾
Investigating Idiomaticity in Word Representations		See Table 1 for random baselines in text samples Relevant for null models.	Paper ▾
 File			Data ▾

To-do

workboard - null models

Assign yourself to tasks you are undertaking then write up the result in a minimal snippet

T _T What	T _T Why	From Meeting #No	Priority M/S/C	Who	Status	Report/ Writeup
Permutation - Word-order shuffle	Randomly permute word order within each text to destroy syntactic structure while preserving the bag of words. Measures syntactic contribution to displacement.	Scientific Evolution #01	M	O...	Not started ▾	N...
POS — replacement (synonym) - Noun synonym substitution	Replace each noun with a synonym (e.g. WordNet/embedding-near est), preserving semantic meaning and token count. Isolates lexical vs. semantic contribution.	Scientific Evolution #01	M	⌘ P...	Not started ▾	N...
Verb synonym substitution	As above, for verbs.	Scientific Evolution #01	S	⌘ P...	Not started ▾	N...
Adjective synonym substitution	As above, for adjectives.	Scientific Evolution #01	S	⌘ P...	Not started ▾	N...
POS — deletion (blank/repeat token) - Noun masking	Replace nouns with a blank/repeated token, holding length constant. An "attention" measure of the noun language factor's influence.	Scientific Evolution #01	M	⌘ P...	Not started ▾	N...
Verb masking	As above, for verbs.	Scientific Evolution #01	S	⌘ P...	Not started ▾	N...
Adjective masking	As above, for adjectives.	Scientific Evolution #01	C	⌘ P...	Not started ▾	N...
Random attack -	Delete N randomly chosen	Scientific	M	⌘ P...	Not started ▾	N...

workboard - null models

Assign yourself to tasks you are undertaking then write up the result in a minimal snippet

Tt What	Tt Why	From Meeting #No	Priority M/S/C	Who	Status	Report/ Writeup
Random N-token deletion	tokens per sentence. Tests sentence-length / token-count influence independent of position.	Evolution #01				
Targeted attack - Head deletion	Delete first N tokens to probe length effect while avoiding mid-sentence meaning breaks.	Scientific Evolution #01	S	⌵ P...	Not started ▾	N...
Tail deletion	Delete last N tokens, same rationale.	Scientific Evolution #01	S	⌵ P...	Not started ▾	N...
Duplication - POS duplication	Repeat adjectives/verbs/nouns N times in situ. Tests sensitivity to local redundancy / repetition.	Scientific Evolution #01	C	⌵ P...	Not started ▾	N...
Extension - Random-sample extension	Append words sampled randomly from the corpus to sentence ends. Length-increase baseline.	Scientific Evolution #01	S	⌵ P...	Not started ▾	N...
Generative extension	Append SciNet/LLM continuation. Tests displacement under semantically-coherent lengthening.	Scientific Evolution #01	S	⌵ P...	Not started ▾	N...
Structural metrics - Sentence length	Compute per-text sentence length and correlate with displacement.	Scientific Evolution #01	M	⌵ P...	Not started ▾	N...
Number of sentences	As above, for sentence count.	Scientific Evolution #01	S	⌵ P...	Not started ▾	N...
PMFI ratio distribution	Compute distribution of PMFI ratio across texts.	Scientific Evolution	C	⌵ P...	Not started ▾	N...

workboard - null models

Assign yourself to tasks you are undertaking then write up the result in a minimal snippet

Tt What	Tt Why	From Meeting #No	Priority M/S/C	Who	Status	Report/ Writeup
		#01				
Alphabetic/numeric - Numeral-to-word substitution	Replace digits with word forms; vary alphabetic:numeric ratio to test numeric-token influence.	Scientific Evolution #01	S	⌘ P...	Not started ▾	N...
Length correlation - len-vs-displacement correlation	Correlate <code>len(m_t)</code> - <code>len(p_t)</code> against <code> m - p </code> (and other combinations) to separate length from semantic displacement.	Scientific Evolution #01	M	lui...	Not started ▾	N...
Domain-specific language - Within-field DSL addition	Inject within-field domain terminology; measure displacement change.	Scientific Evolution #01	S	⌘ P...	Not started ▾	N...
Out-of-field DSL addition	Inject out-of-field DSL as contrast.	Scientific Evolution #01	S	⌘ P...	Not started ▾	N...
Norm-based variation - Concreteness-graded displacement	Vary displacement by concreteness norms (37,058-word coverage); test concrete vs. abstract sensitivity.	Scientific Evolution #01	C	⌘ P...	Not started ▾	N...
Specificity-graded displacement	As above, using specificity norms (8,500-word coverage).	Scientific Evolution #01	C	⌘ P...	Not started ▾	N...
Descriptive:noun ratio	Compare counts of adjectives/descriptive words vs. nouns against displacement.	Scientific Evolution #01	C	⌘ P...	Not started ▾	N...
Comparing embedding models	How generic like ada-002 compares to fine-tuned, like specter2?			⌘ P...	Not started ▾	A...

Meeting Minutes

Jun 11, 2026 | 📅 Science of Science

Attendees: Luiz/Owen/Ronaldo/Chico

Notes

- Research partitioned into a Universal Schema of “A (P)roblem approached by (M)ethod gave (F)inding with (I)nterpretation” - identified via LLM (robustness checks required?)
- PMFI separated into Multilayer Unit Embedding Space (w2v/semantic layers)
- Trajectories of papers across layers as analogue to exploitation and exploration of the research space and scientific knowledge landscape
- Measure of creativity based on divergence of trajectories from A) each other B) embedding space origin C) domain mean trajectory - essentially density based, see [Charting Mobility Patterns](#) for motivation here
- Measure of creativity based on entropy of trajectories of A) individuals B) Domains C) Venues
- Procrustes Alignment for a Longitudinal Study - taking multiple static slices and seeing the change in embedding space and the tolerance of trajectories over time.
- Sensitivity parameters for the influence of each layer on research impact (citation as impact target?)
- Simplicial Complexes for the relations between PMFI and potential Higher Order Interactions (HOI) - e.g. influence of problem on other three layers contains less signal/explanation than influence of problem+method on other two layers vs has completely different signal/explanation than typical average problem+method connections in this area
- Semantic Distance in Embedding Space between a paper's Problem and it's Interpretation that's been projected onto the Problem embedding space - distances represent change in framing, e.g. Computer Science problem to Logic interpretation

Action items

- ☒ ~~Planning/Prototyping Meeting with just Luiz and Owen~~
- ☒ ~~Set Goals and Tasks - Null Models as First Stage~~

Jun 12, 2026 | 📅 Scientific Evolution #01

Attendees: Luiz/Owen

Agenda

- Goals/Stages/Milestones Low Hanging Fruit
- Work Breakdown
- Prototyping Data (~Computer Science)
- Data Exploration and Analysis(?)
- Hypotheses/Expectations

Notes

- Null tests for real data:
 - Permutation null - shuffling word order to remove syntactic structure
 - Nouns/Verbs/Adjectives - with replacement (same semantic meaning for word, but use a synonym), ensuring text length. With deletion - replacing Vs or Ns with a blank token/repeated token - a sort of attention measure for language factor influence
 - Random attack - N-deleted tokens chosen randomly from sentence to see sentence length influence
 - Targeted attack - tail deletion/head deletion of N tokens to avoid breaking sentence meanings
 - Duplication - repeating adjectives/verbs/nouns N times in the same area
 - Extension - adding words on the end either randomly sampled or generated by SciNet/LLM continuation etc.
 - Sentence Length, Number of Sentences, Distribution of PMFI ratio
 - Ratio alphabetic to numeric - replacing numbers with their word versions
 - Correlation between $\text{len}(m_t) - \text{len}(p_t)$ and $||m - p||$ (and other combinations)
 - Addition of within field vs without field domain specific language (DSL)
 - Variation of displacement based on Norms - concreteness/specificity with 37058/8500 coverage respectively of english words. Number of adjectives/descriptive words compared to nouns etc.
- Hypotheses 1-3 have essentially been proven already except for quantum-chaos data subset - need to see why it fails here.
- (WIP) Next Paper: [Modeling Scientific Discovery as Discourse Trajectories in Epistemic Landscapes](#)
- (WIP) Math Formulation: [Towards a mathematical theory of science](#)
- Prototyping data is on OpenALex topics for null models

Action items

- ☒ ~~Owen access to data~~
- ☒ ~~Writeup minutes~~
- ☒ ~~Assign null models for validating and writeup [tasks](#) based on priority~~
- ☐ Brainstorm - quantum_chaos is an outlier for hypotheses, why?
- ☐ Forwarded to next Agenda - Null results, Data Exploration and Analysis, Next Stages/Milestones